Rotary Screw Air Compressor (AIR-COMP) Equipment Manual and SOP

VINDHYA PRECISION WORKS
MAINTENANCE ENGINEERING DEPARTMENT
STANDARD OPERATING PROCEDURE & EQUIPMENT MANUAL
ASSET CLASS: AIR-COMP (ROTARY SCREW AIR COMPRESSOR)
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1.0 PURPOSE AND SCOPE

1.1 This document defines the operating parameters, preventive maintenance schedules, fault diagnosis protocols, and spare parts provisioning for asset class AIR-COMP (Rotary Screw Air Compressors) installed at Vindhya Precision Works.

1.2 Compliance with this manual is mandatory for all maintenance technicians, reliability engineers, and production supervisors operating within the Compressor House.

2.0 RESPONSIBLE TRADES AND HAZARDOUS ENERGY ISOLATION

2.1 Responsible Trades
* Primary Trade: Mechanical.
* Supporting Trade: Instrumentation.

2.2 Hazardous Energy Sources
Prior to the commencement of any maintenance, inspection, or non-routine intervention, technicians shall isolate the following energy sources in accordance with Vindhya Precision Works Lockout/Tagout (LOTO) Procedure VPW-LOTO-001:
1. Electrical: 415V AC, 3-Phase, 50Hz supply via Main Incoming Feeder Breaker.
2. Pneumatic: Compressed Air System via Upstream and Downstream Isolation Valves with downstream auto-venting line to zero energy state.

3.0 NORMAL OPERATING RANGES AND CONDITION MONITORING THRESHOLDS

3.1 Operating Parameters

Equipment operating within acceptable baseline health parameters at Vindhya Precision Works shall register within the following narrow bands:

* Healthy Vibration: 1.2 to 4.0 mm/s RMS (measured at airend bearing housings, horizontal and vertical axes).
* Healthy Surface Temperature: 38 °C to 66 °C (measured via IR thermography and PT100 elements on airend discharge and air-oil separator vessel).
* Nominal Operating Pressure: 7.5 bar.

3.2 Condition Monitoring Alarm and Trip Limits

The plant condition-monitoring system (CMS) is programmed with hard limits. Any violation of these thresholds shall initiate automated logging, operator warnings, or emergency trips as specified below:

Parameter	Baseline / Normal	Warning Setpoint	Trip Setpoint	Action on Trip
Vibration (RMS)	1.2 – 4.0 mm/s	6.3 mm/s	9.5 mm/s	Immediate motor coast-to-stop; lockout auto-restart
Surface Temperature	38 °C – 66 °C	88 °C	102 °C	Immediate load-solenoid de-energisation and shutdown
Operating Pressure	7.5 bar nominal	6.2 bar (Low Pressure)	N/A	Generates SCADA warning; does not trip
Motor Current	110 A nominal	145 A	165 A	Immediate breaker trip

4.0 PREVENTIVE MAINTENANCE SCHEDULE BY RUNNING HOURS

4.1 Base Preventive Maintenance Interval
The base preventive maintenance interval for asset class AIR-COMP is strictly 2000 running hours (± 50 tolerance hours). Maintenance must be executed by certified Mechanical and Instrumentation technicians.

4.2 Schedule of Tasks

* 2,000 Running Hours (Level 1 - Minor Service):
* Inspect and tension My belt (VPW-P-00036) to deflection specification of 4.5 mm under 35 N force.
* Replace Road seal (VPW-P-00001).
* Test Give valve (VPW-P-00034) for proper cycling and leakage.
* Inspect Per contactor (VPW-P-00019) and Eight contactor (VPW-P-00050) for contact pitting; measure contact resistance.
* Verify calibration of Time sensor (VPW-P-00106).

* 4,000 Running Hours (Level 2 - Intermediate Service):
* Execute all Level 1 tasks.
* Inspect Sell bearing (VPW-P-00003) condition via acoustic emission and thermal profiling.
* Replace Give valve (VPW-P-00034).
* Replace Such contactor (VPW-P-00075).

* 8,000 Running Hours (Level 3 - Major Overhaul Service):
* Execute all Level 1 and Level 2 tasks.
* Replace Sell bearing (VPW-P-00003).
* Replace Per contactor (VPW-P-00019) and Eight contactor (VPW-P-00050) regardless of visual condition.
* Perform complete alignment check of drive motor to airend.

5.0 FAULT-CODE TABLE AND DIAGNOSTIC PROSE

5.1 Technicians shall use the following fault descriptions, probable root causes, and corrective actions for troubleshooting asset class AIR-COMP:

1. Fault Code F-101: High Air Discharge Temperature
* *Probable Cause:* Low compressor lubricant level, fouled air-oil cooler matrix, or failing Time sensor (VPW-P-00106) providing false readings.
* *Corrective Action:* Check oil level in separator vessel and top up if required. Clean external cooler fins using compressed air. If temperature persists above 88 °C, check calibration of Time sensor (VPW-P-00106) and replace if output drifts by >2%.

2. Fault Code F-102: Excessive Vibration Warning (6.3 mm/s RMS)
* *Probable Cause:* Loose My belt (VPW-P-00036) causing pulley resonance, or initial degradation of Sell bearing (VPW-P-00003).
* *Corrective Action:* Isolate electrical energy, inspect My belt (VPW-P-00036) tension and alignment, and re-torque mounting bolts. If vibration remains elevated, inspect Sell bearing (VPW-P-00003) for spalling.

3. Fault Code F-103: Low Operating Pressure (< 6.2 bar)
* *Probable Cause:* Stuck or sluggish Give valve (VPW-P-00034) failing to close fully during loading cycle, or severe downstream compressed air system leakage.
* *Corrective Action:* Isolate pneumatic energy, inspect Give valve (VPW-P-00034) for internal debris or diaphragm rupture. Clean or replace Give valve (VPW-P-00034). Conduct ultrasonic leak detection on header network.

4. Fault Code F-104: Main Motor Overcurrent Trip (165 A)
* *Probable Cause:* Airend rotor binding due to Sell bearing (VPW-P-00003) failure, or phase imbalance across Per contactor (VPW-P-00019).
* *Corrective Action:* Perform manual shaft rotation check of airend (unpowered). Check contact surfaces of Per contactor (VPW-P-00019) for welding or high resistance. Replace Sell bearing (VPW-P-00003) if mechanical binding is identified.

5. Fault Code F-105: Control Circuit Failure / Starter Drop-Out
* *Probable Cause:* Burnout or mechanical binding in Eight contactor (VPW-P-00050) or auxiliary coil failure in Such contactor (VPW-P-00075).
* *Corrective Action:* Isolate electrical energy. Measure coil resistance of Eight contactor (VPW-P-00050) and Such contactor (VPW-P-00075). Replace faulty contactor units.

6. Fault Code F-106: Oil Leakage at Shaft Interface
* *Probable Cause:* Degradation and weeping of Road seal (VPW-P-00001) due to thermal cycling beyond design life.
* *Corrective Action:* Isolate electrical and pneumatic energy sources. Disassemble drive coupling, extract worn Road seal (VPW-P-00001), inspect shaft sleeve for scoring, and install new Road seal (VPW-P-00001) using dedicated installation arbor.

6.0 CONSUMABLE AND SPARE PARTS LIST

6.1 Technicians shall only use inventory items matching the exact part numbers and descriptions recorded in the Vindhya Precision Works enterprise asset management system for asset class AIR-COMP. No substitutions are permitted.

* `VPW-P-00001` — Road seal
* `VPW-P-00003` — Sell bearing
* `VPW-P-00019` — Per contactor
* `VPW-P-00034` — Give valve
* `VPW-P-00036` — My belt
* `VPW-P-00050` — Eight contactor
* `VPW-P-00075` — Such contactor
* `VPW-P-00106` — Time sensor

7.0 TORQUE SPECIFICATIONS

7.1 Maintenance personnel shall apply calibrated torque wrenches to all fasteners associated with asset class AIR-COMP in accordance with the following schedule:

* Airend Discharge Flange Bolts (M12 Gr 8.8): 85 Nm (± 4 Nm)
* Motor Foot Mounting Bolts (M16 Gr 8.8): 195 Nm (± 10 Nm)
* Air-Oil Separator Cover Nuts (M10 Gr 8.8): 50 Nm (± 2 Nm)
* Electrical Power Terminals (Contactors VPW-P-00019, VPW-P-00050, VPW-P-00075): 14 Nm (± 1 Nm)

8.0 SAFETY-CRITICAL CORRECTIVE ACTIONS AND PERMIT REQUIREMENTS

8.1 The following corrective actions are designated as Safety-Critical. They cannot be performed without an approved Work Permit (Permit to Work - PTW) and complete execution of Plant Lockout/Tagout (LOTO) protocols as detailed in section 2.0:

1. Replacement or internal overhaul of the Sell bearing (`VPW-P-00003`).
2. Replacement or servicing of the Road seal (`VPW-P-00001`) involving shaft disassembly.
3. Any intervention on electrical contactors (`VPW-P-00019`, `VPW-P-00050`, `VPW-P-00075`) or internal wiring.
4. Opening, inspection, or resealing of the air-oil separator pressure vessel or associated Give valve (`VPW-P-00034`) pipework.
5. Tensioning or replacement of My belt (`VPW-P-00036`) involving guard removal and proximity to rotating transmission shafts.

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